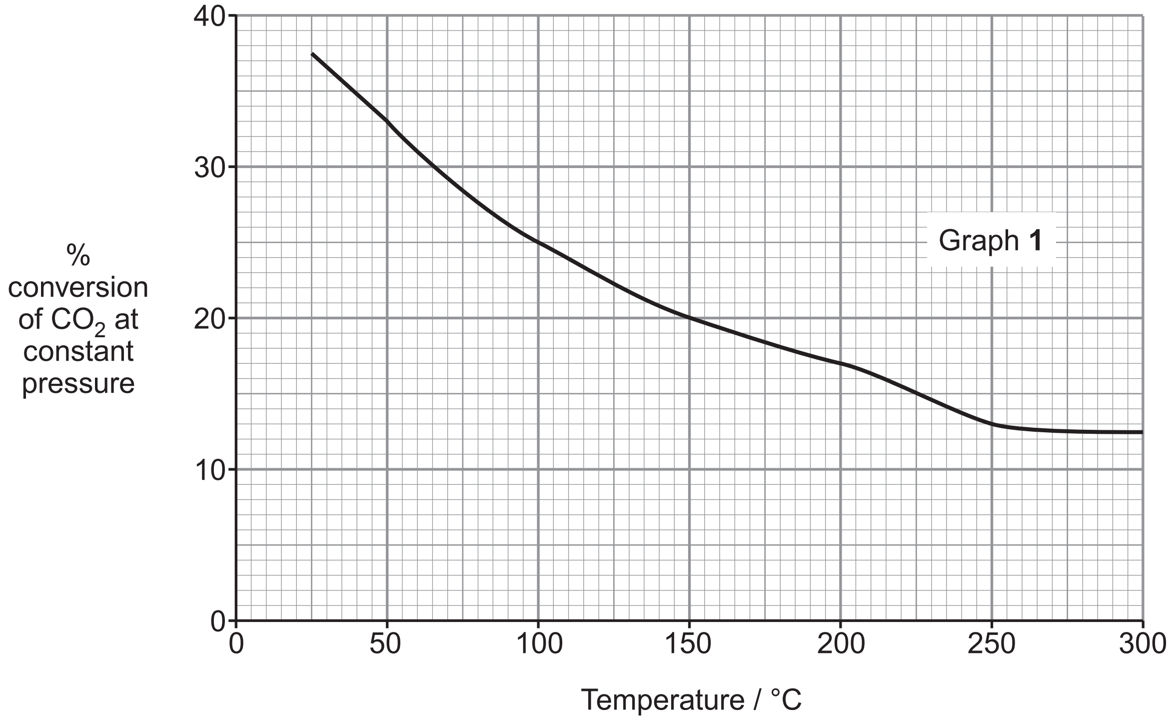
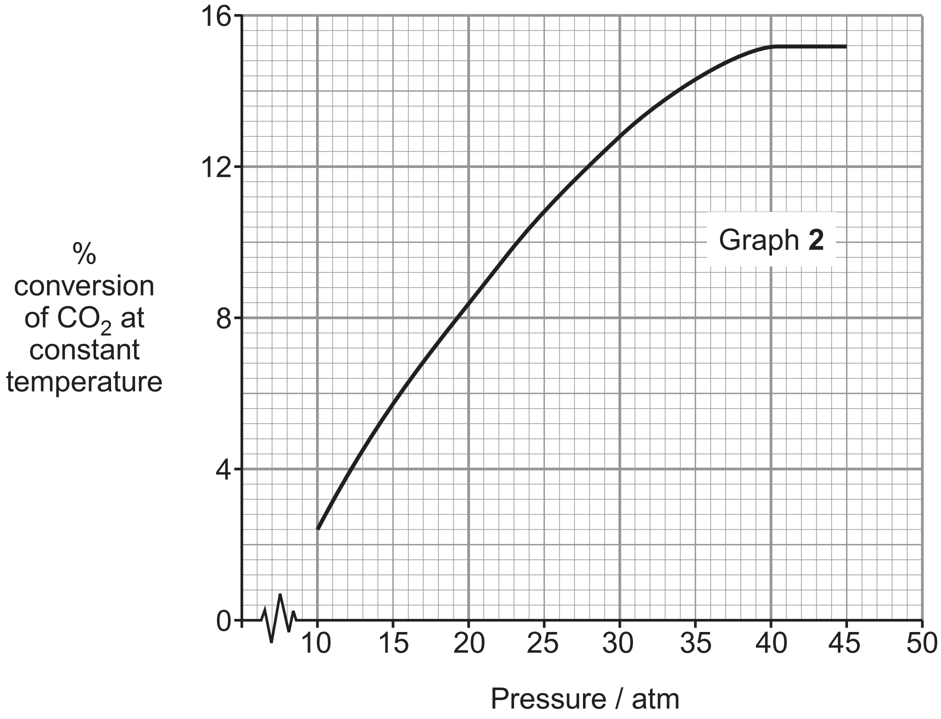
12. (a) There are several stages in the industrial production of methanol from methane.

The final stage involves an equilibrium where carbon dioxide is converted into methanol. The graphs below show the percentage conversion of carbon dioxide into methanol at **equilibrium** under different conditions of temperature and pressure.





Examiner only

Equilibrium data is not the only information used when choosing the optimum temperature and pressure for the reaction.

Use the graphs and your knowledge to

- give as much information as possible about the reaction
- suggest optimum conditions for the reaction and indicate what further information is needed to make this judgment

Explain your reasoning throughout. [6 QER]

Examiner only

(b) The hydrogenation of carbon monoxide is a different method of producing methanol.

$$CO(g) \quad + \quad 2H_2(g) \rightleftharpoons CH_3OH(g)$$

(i) Write an expression for the equilibrium constant, K_c, and give its unit, if any. [2]

K_c =

Unit ....

(ii) The equilibrium constant, K_c, has a numerical value of 0.350 at a temperature of 350 °C.

If the equilibrium concentration of carbon monoxide is 0.725 mol dm^{-3} and that of hydrogen is 0.850 mol dm^{-3}, calculate the equilibrium concentration of methanol at this temperature. [2]

Concentration = ... mol dm^{-3}

Examiner only

(c) Carbon monoxide is used in the reduction of iron(III) oxide in the blast furnace.

$$Fe_2O_3(s) \quad + \quad 3CO(g) \longrightarrow 2Fe(l) \quad + \quad 3CO_2(g)$$

(i) Calculate the atom economy for the formation of iron in this reaction. Give your answer to an **appropriate** number of significant figures. [3]

Atom economy = ... %

(ii) Calculate the volume, in m^3, of carbon dioxide that would be produced if 20 tonnes of iron(III) oxide were reduced at a temperature of 1100 °C and a pressure of 1 atm. [4]

Volume = ... m^3

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